

Random common 80k external test: raw vs catastrophe-filtered
Architecture averages across seeds ($\pm$ sample standard deviation)

Architecture	Metric	Seeds	Raw mean $\pm$ seed SD	Filtered mean $\pm$ seed SD	Removed
L3N4	Energy MAE	6	0.219 $\pm$ 0.027	0.212 $\pm$ 0.015	7
L3N4	Energy MSE	6	17.275 $\pm$ 41.359	0.133 $\pm$ 0.016	7
L3N4	Force MAE	6	0.898 $\pm$ 0.726	0.603 $\pm$ 0.027	9
L3N4	Force RMSE	6	108.529 $\pm$ 258.380	1.328 $\pm$ 0.045	10
L3N5	Energy MAE	8	0.227 $\pm$ 0.030	0.225 $\pm$ 0.031	9
L3N5	Energy MSE	8	1.030 $\pm$ 2.419	0.150 $\pm$ 0.036	9
L3N5	Force MAE	8	0.684 $\pm$ 0.118	0.633 $\pm$ 0.048	13
L3N5	Force RMSE	8	21.675 $\pm$ 52.367	1.382 $\pm$ 0.098	13
L4N3	Energy MAE	6	0.213 $\pm$ 0.004	0.213 $\pm$ 0.004	7
L4N3	Energy MSE	6	0.143 $\pm$ 0.019	0.135 $\pm$ 0.004	7
L4N3	Force MAE	6	0.624 $\pm$ 0.014	0.623 $\pm$ 0.013	7
L4N3	Force RMSE	6	1.545 $\pm$ 0.329	1.512 $\pm$ 0.338	8
L4N4	Energy MAE	4	0.184 $\pm$ 0.003	0.184 $\pm$ 0.003	4
L4N4	Energy MSE	4	0.130 $\pm$ 0.049	0.104 $\pm$ 0.004	4
L4N4	Force MAE	4	0.553 $\pm$ 0.010	0.551 $\pm$ 0.011	4
L4N4	Force RMSE	4	1.615 $\pm$ 0.587	1.244 $\pm$ 0.026	4